# Problem Statement No. 1 - Research Agent

*Technology: IBM Cloud Lite services / IBM Granite (mandatory)*

## The Challenge

A Research Agent is an AI system designed to assist with academic and scientific research tasks. It can autonomously search for literature, summarize papers, and organize references. Using natural language processing, it understands research questions and retrieves relevant information. The agent can generate reports, suggest hypotheses, and even draft sections of research papers. It saves time by automating repetitive tasks like citation management and data extraction. Research Agents enhance efficiency, accuracy, and innovation in both academic and industrial R&D.

## Our Solution

We built an autonomous Research Agent on IBM Granite (watsonx.ai, IBM Cloud Lite). From a single natural-language research question it plans queries, retrieves papers from arXiv and OpenAlex, summarizes each paper into structured fields, writes a cited literature review, proposes testable hypotheses, drafts paper sections and exports APA/BibTeX references.

| Requirement | How it is addressed |
|---|---|
| Search literature | Granite-planned queries; arXiv + OpenAlex retrieval with de-duplication |
| Summarize papers | Granite extracts objective, method, findings, limitations, datasets/metrics |
| Organize references | APA list and BibTeX export; optional IBM COS storage |
| Generate reports | Cited literature review exported as Markdown |
| Suggest hypotheses | Gap-grounded hypotheses with rationale and validation plan |
| Draft paper sections | Abstract, Introduction and Methodology drafts |
| Data extraction | Structured JSON extraction per paper |
| IBM technology | watsonx.ai Runtime + Granite, COS Lite, Code Engine |